

Level 3 — Hard **Median (H)** — functionsandpointers

Write a function `double findMedianSortedArrays(int *arr1, int n1, int *arr2, int n2)` that finds the median of two sorted arrays using pointers.

Sample Input:

Enter size of first array: 3

Enter elements of first array: 1 3 5

Enter size of second array: 3

Enter elements of second array: 2 4 6

Sample Output:

Median is 3.5

Your Code

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 double findMedianSortedArrays(int *arr1, int n1, int *arr2, int n2) {
5     int total = n1 + n2;
6     int *merged = (int *)malloc(total * sizeof(int));
7     int i = 0, j = 0, k = 0;
8
9     // Merge the two sorted arrays using pointers
10    while (i < n1 && j < n2) {
11        if (*(arr1 + i) <= *(arr2 + j)) {
12            *(merged + k) = *(arr1 + i);
13            i++;
14        } else {
15            *(merged + k) = *(arr2 + j);
16            j++;
17        }
18        k++;
19    }
20    while (i < n1) {
21        *(merged + k) = *(arr1 + i);
22        i++;
23        k++;
24    }
25    while (j < n2) {
26        *(merged + k) = *(arr2 + j);
27        j++;
28        k++;
29    }
30    int mid = total / 2;
31    if (total % 2 == 0) {
32        return (*(merged + mid - 1) + *(merged + mid)) / 2.0;
33    } else {
34        return *(merged + mid);
35    }
36}
```

Program Input (stdin)

Enter input if required...

Output

```
Enter size of first array: Enter elements
of first array: Enter size of second
array: Enter elements of second array:
Median is 0.0
```

Run

Save

Level 3 — Hard **gcd (H)** — function

Write a function `int gcd(int a, int b)` that computes the greatest common divisor of two numbers a and b.

Sample Input:

Enter two numbers: 56 98

Sample Output:

GCD of 56 and 98 is 14

Your Code

```
1 #include <stdio.h>
2
3 int gcd(int a, int b) {
4     while (b != 0) {
5         int temp = b;
6         b = a % b;
7         a = temp;
8     }
9     return a;
10 }
11
12 int main() {
13     int a, b;
14     printf("Enter two numbers: ");
15     scanf("%d %d", &a, &b);
16
17     printf("GCD of %d and %d is %d\n", a, b, gcd(a, b));
18
19     return 0;
20 }
```

Program Input (stdin)

Enter input if required...

Output

```
Enter two numbers: GCD of 45 and 76 is 1
```

Run

Save